

Machine Learning - *Temporal* models (69152) **RNNs, Transformers, ...**

Master in Robotics, Graphics and Computer Vision
Ana C. Murillo



Universidad
Zaragoza

Today

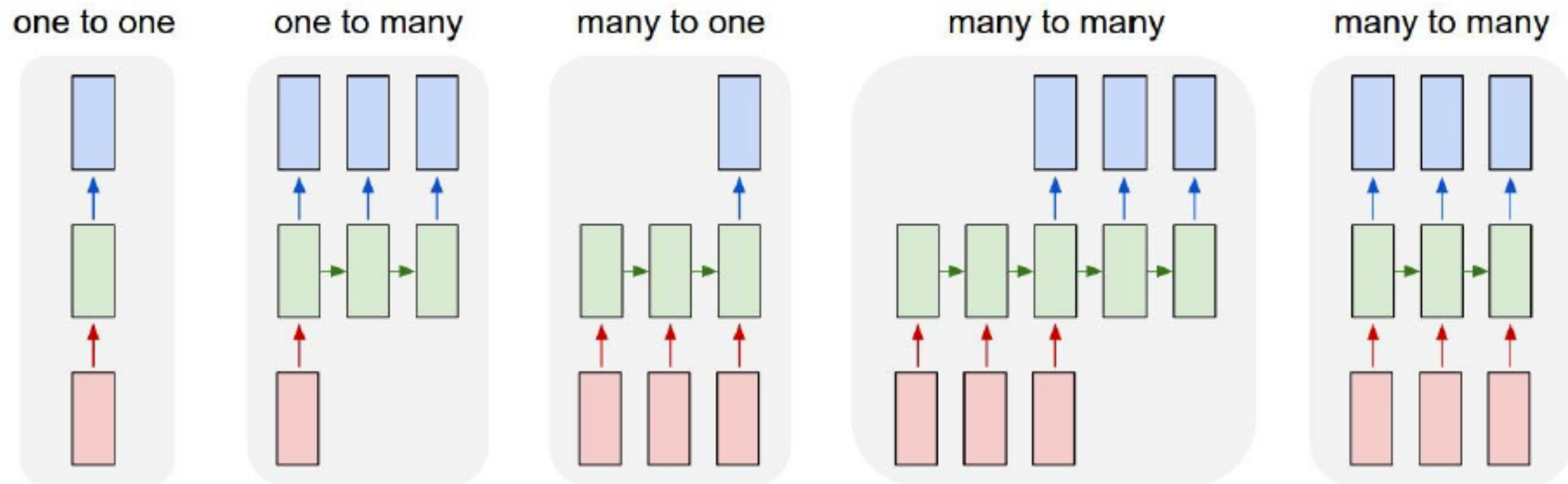
- RecursiveNN
- Resources:
 - <http://cs231n.stanford.edu>
 - <http://karpathy.github.io/2015/05/21/rnn-effectiveness/>
—> toy examples of RNN in python
 - Deep Learning book. Chapter 10.
www.deeplearningbook.org

Deep Learning && ...

- Time? Sequential Data?

Deep Learning & Sequential data

- Video and Speech/Text, etc? —> **Recurrent NN**



Fei-Fei Li & Justin Johnson & Serena Yeung
<http://cs231n.stanford.edu>

- can handle **non-fixed length input/output**

Recurrent Neural Networks

- repeat (as many input/output/steps as needed)
- **share the weights** across iterations

one to one

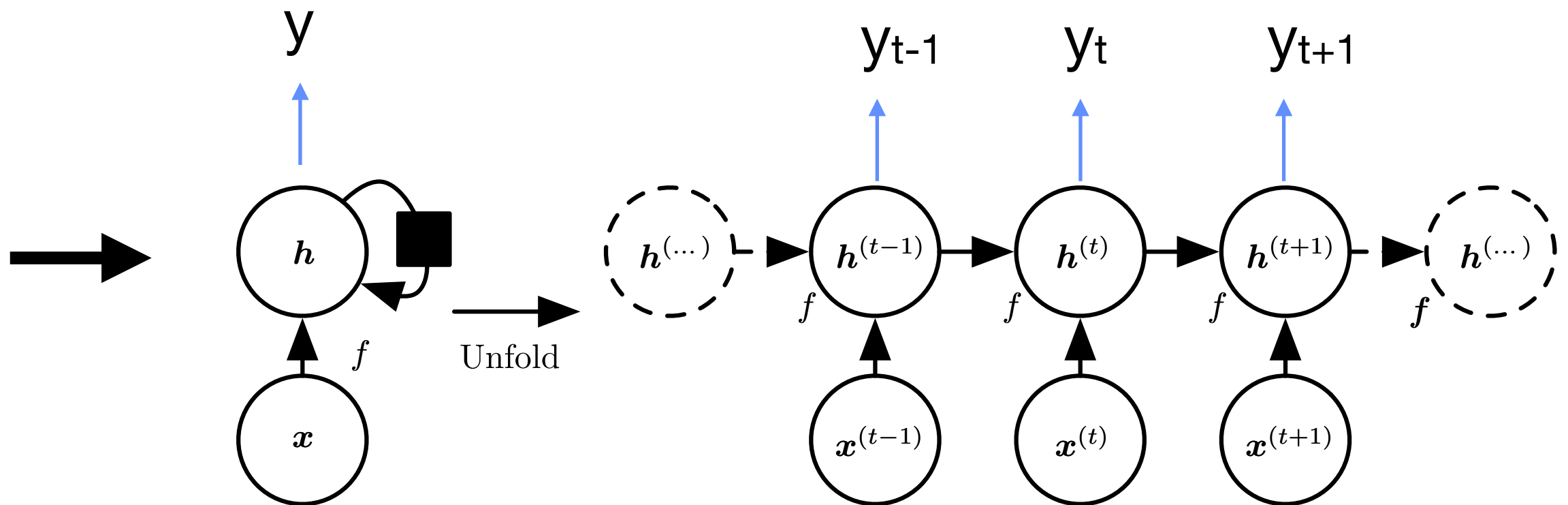
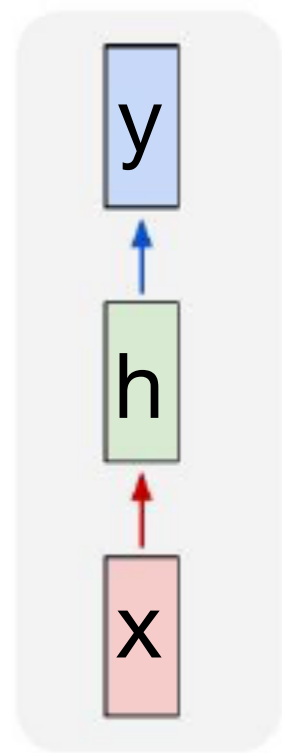


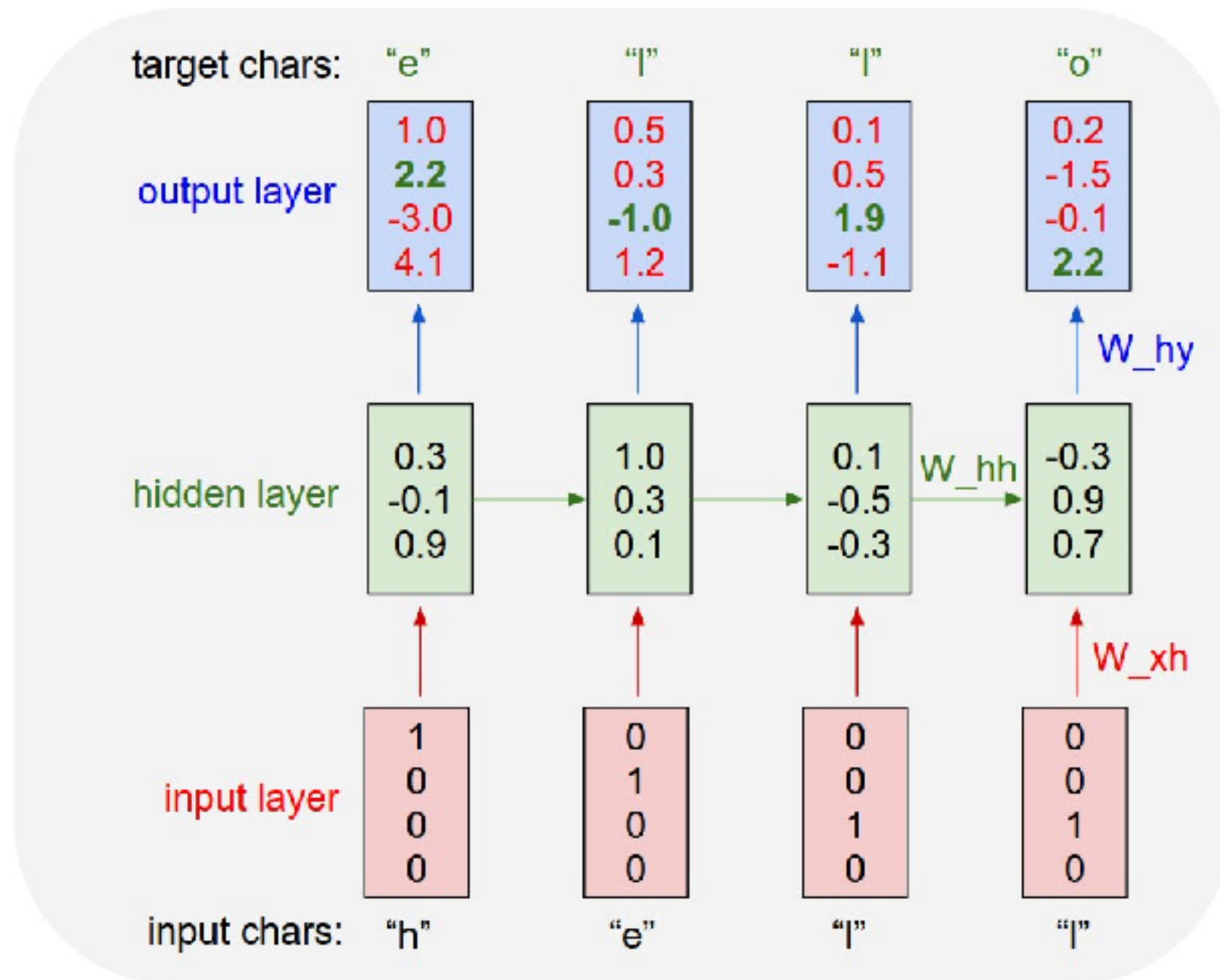
Figure 10.2 - Deep Learning book. Ian Goodfellow

Recurrent Neural Networks

Main ingredients on a toy example:

- time: t
- hidden layer (s): h
- “vocabulary”: $\{h, e, l, o\}$
- weights: $\mathbf{W}_{xh}$, $\mathbf{W}_{hh}$, $\mathbf{W}_{hy}$

$$h_t = \tanh(\mathbf{W}_{hh}h_{t-1} + \mathbf{W}_{xh}x_t)$$
$$y_t = \mathbf{W}_{hy}h_t$$



Recurrent Neural Networks

Interesting examples from a *vainilla* version

- text generator (Shakespeare, Wikipedia page, ...)
- code generator! (LaTeX, C, ...)

<http://karpathy.github.io/2015/05/21/rnn-effectiveness/>

Recurrent Neural Networks

Neuron that highlights URLs

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input sequence

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5 most likely characters for each sequence input char

<http://karpathy.github.io/2015/05/21/rnn-effectiveness/>

Recurrent Neural Networks

Neuron that highlights URLs

input sequence

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5 most likely
characters
for each
sequence
input char

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Recurrent Neural Networks

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Recurrent Neural Networks

But ...

RNN vainilla version does not work that well in practice

Recurrent Neural Networks

- **LSTM networks** - Long short-term memory cells
 - Very frequent type of RNN
 - more sophisticated hidden function than basic RNN
 - capable of learning long-term dependencies

$$h_t = \tanh \left(W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix} \right) \quad \longrightarrow \quad \begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$
$$c_t = f \odot c_{t-1} + i \odot g$$
$$h_t = o \odot \tanh(c_t)$$

Recurrent Neural Networks

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$$h_t = \tanh \left(W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix} \right)$$

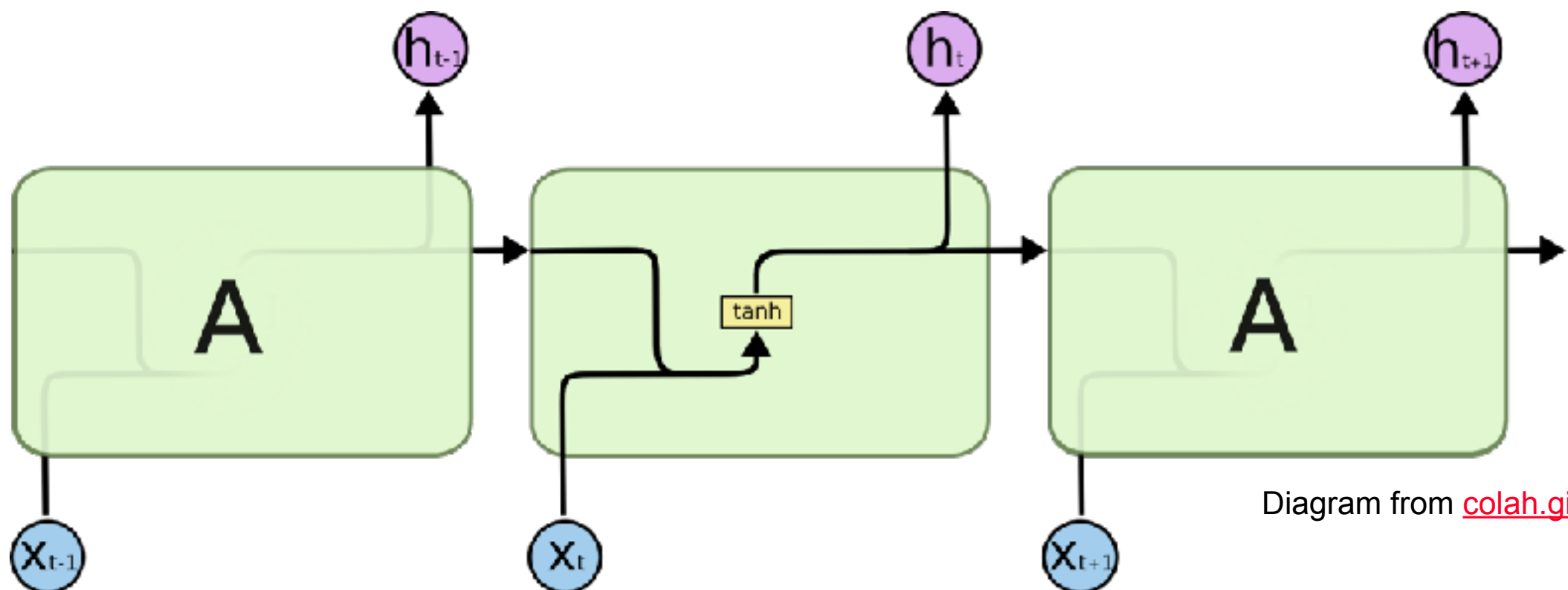


Diagram from colah.github.io

Recurrent Neural Networks

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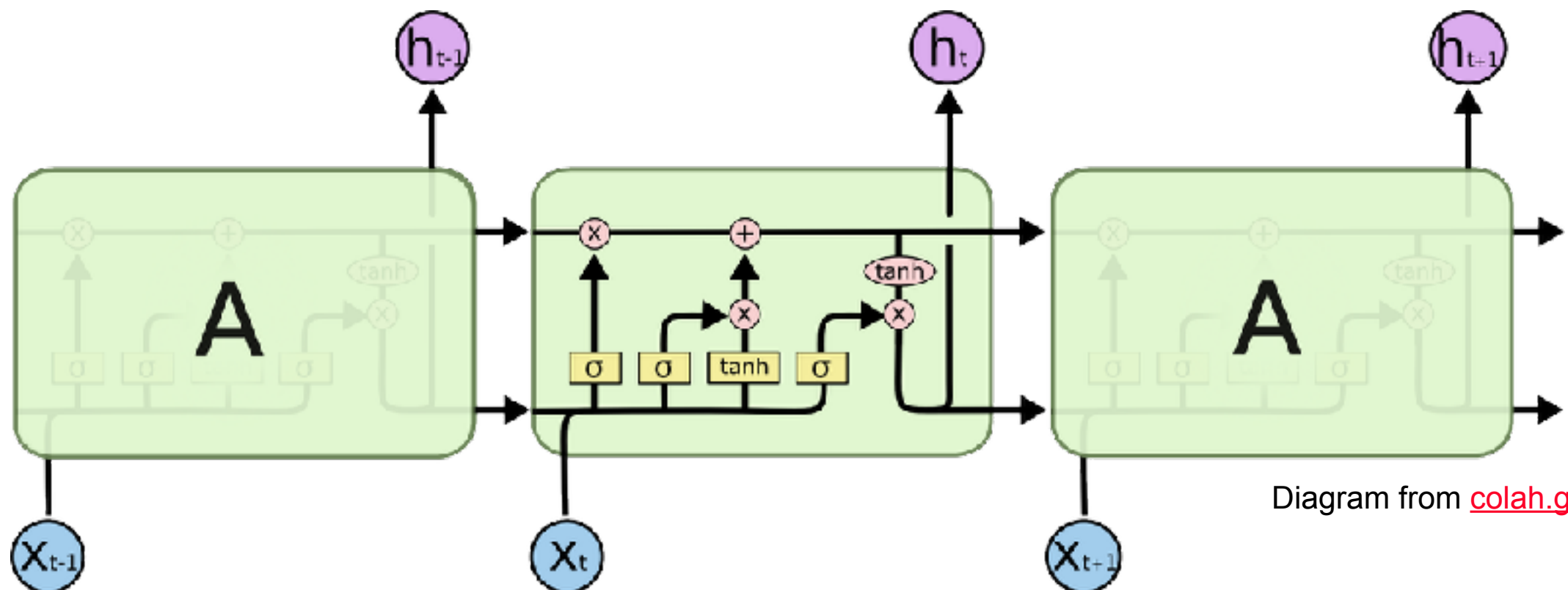
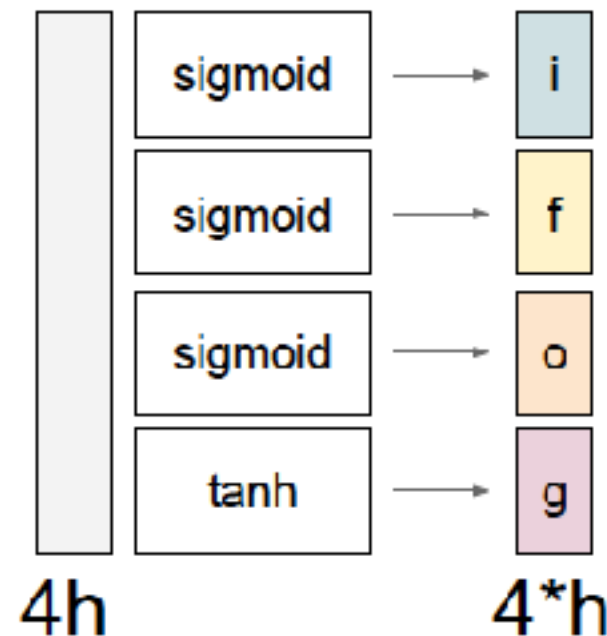


Diagram from colah.github.io

Recurrent Neural Networks

- **LSTM networks** - Long short-term memory cells

- **f: forget** gate.
Erase cell or not?
- **i: input** gate.
Write cell or not?
- **g: Cell Gate**.
How much to write to cell
- **o: Output** gate.
How much to reveal cell value

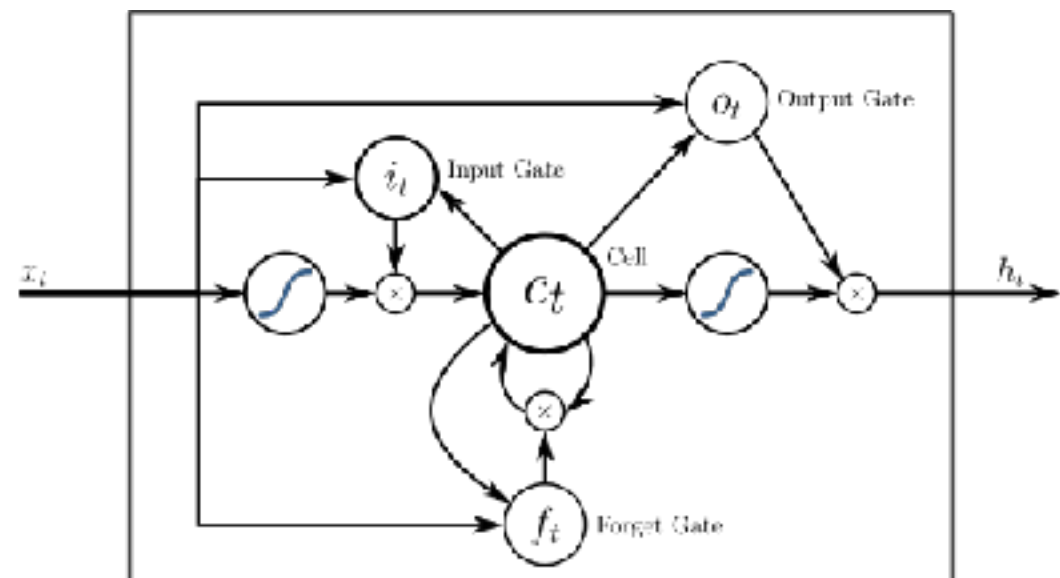


Fei-Fei Li & Justin Johnson & Serena Yeung
<http://cs231n.stanford.edu>

$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

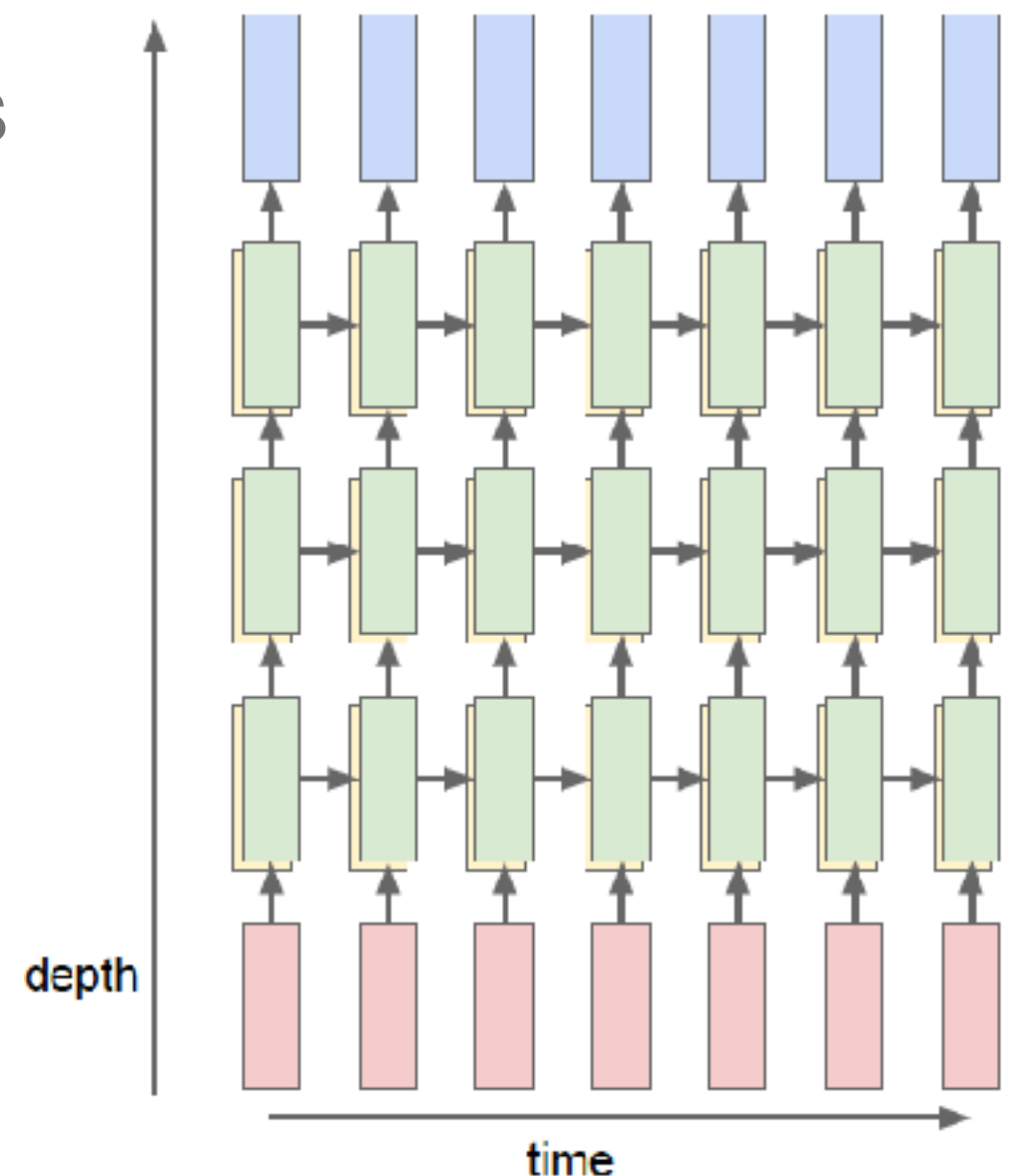
$$c_t = f \odot c_{t-1} + i \odot g$$

$$h_t = o \odot \tanh(c_t)$$



Recurrent Neural Networks

- Many successful examples and variations:
 - Many variations of LSTM cells
 - Multi-layer RNN

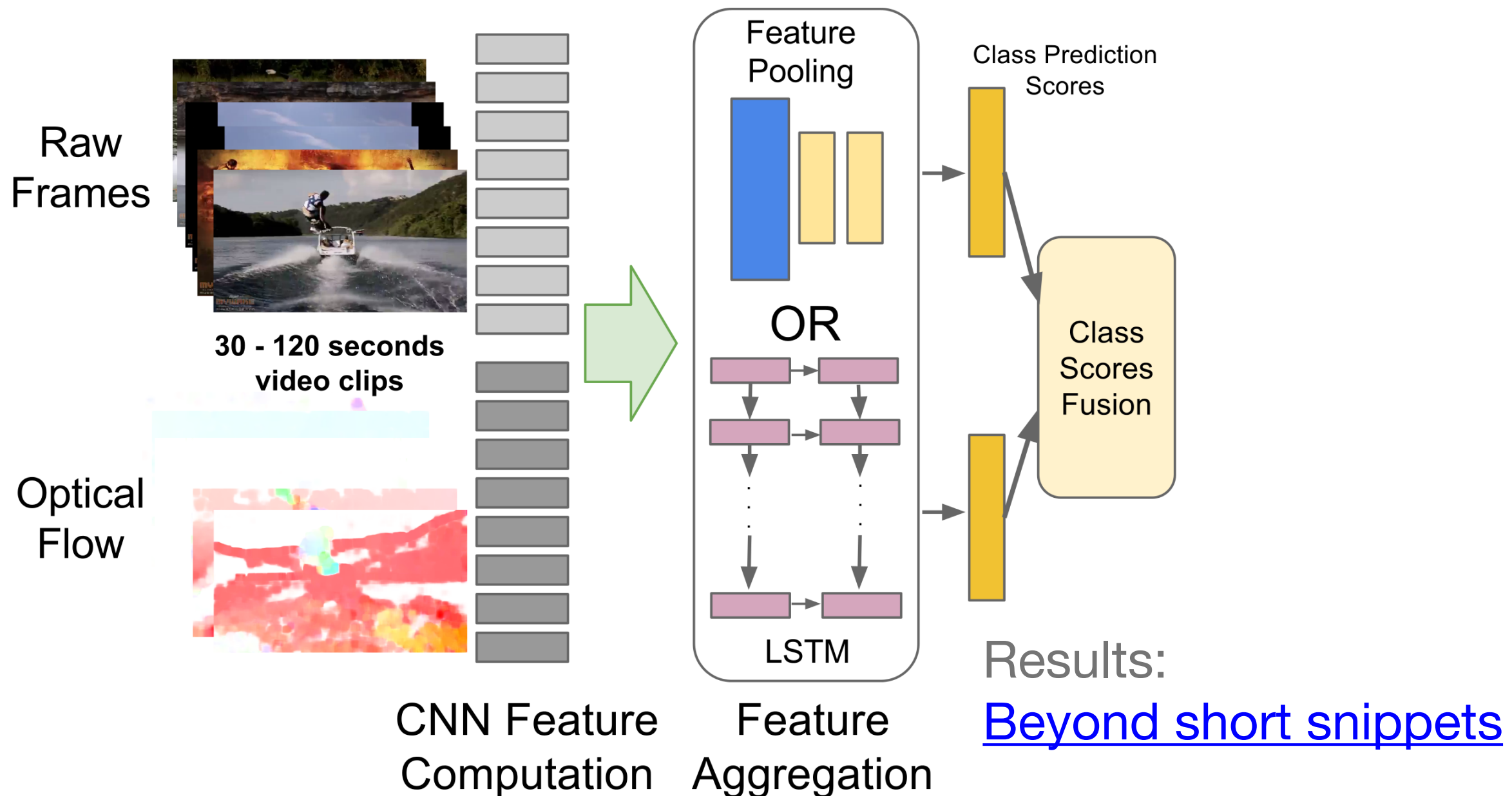


Recurrent Neural Networks

- Besides “pure” RNN examples
- plenty of combinations of RNN with others

Recursive NN examples

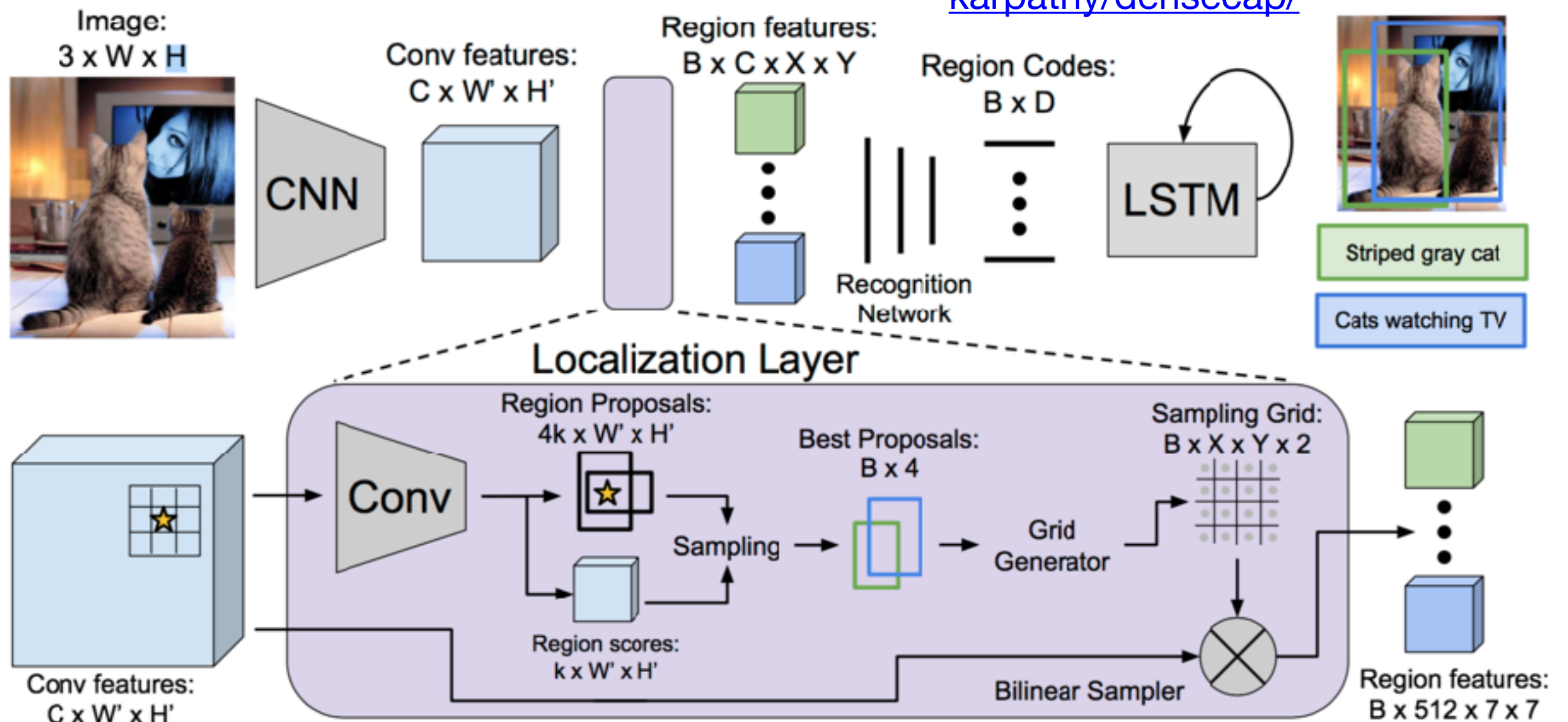
- **Action Classification.** Video and Flow (temporal data)



Recursive NN examples

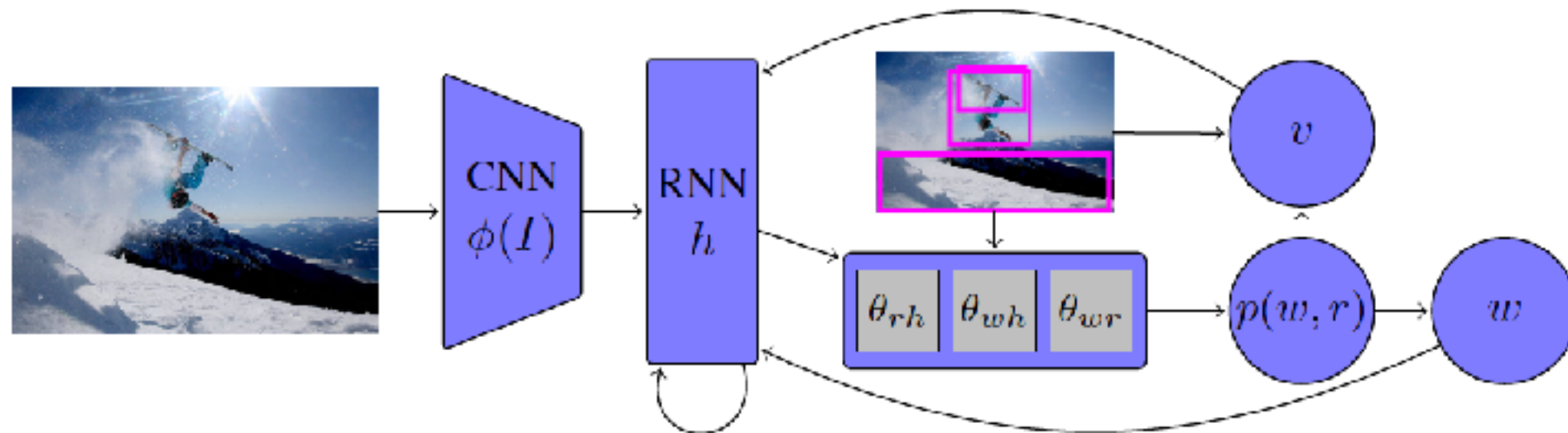
- **Multimodal problems.** Image and Text. Dense Captions (other types of sequential data)

Results: <http://cs.stanford.edu/people/karpathy/densecap/>



Recursive NN examples

- **Multimodal problems.** Image and Text. Captioning & Attention (other types of sequential data)



RNN & CuDNN example

Exercise in class: we will implement a simple RNN.

Build a text file of similar size to this one (200K words):

<https://storage.googleapis.com/download.tensorflow.org/data/shakespeare.txt>

of certain type of text/speech

e.g., speech from someone; lyrics of certain type; programming language; ... some more examples here:

<http://karpathy.github.io/2015/05/21/rnn-effectiveness/>)

RNN & CuDNN example/exercise

- Use your own text file of similar size to this one (200K words):
<https://storage.googleapis.com/download.tensorflow.org/data/shakespeare.txt> of certain type of text/speech (e.g., speech from someone; lyrics of certain type; programming language; ... some more examples here: <http://karpathy.github.io/2015/05/21/rnn-effectiveness/>)
- **Run Colab example** to implement a simple RNN using your text style data:
https://colab.research.google.com/drive/1Qjcip3iErC_UyKEL62DwO_S2GBWKrxMo
(extracted from https://www.tensorflow.org/tutorials/text/text_generation)
 1. Read and understand the current example
 2. Re-run with your own data

Next

- RNN exercise
- **QUESTIONNAIRES** - <https://encuestas.unizar.es/english-questionnaires>
- Other *temporal* models
 - TCN
 - Transformers